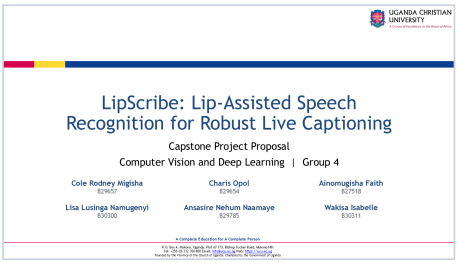
LipScribe: talking points per slide

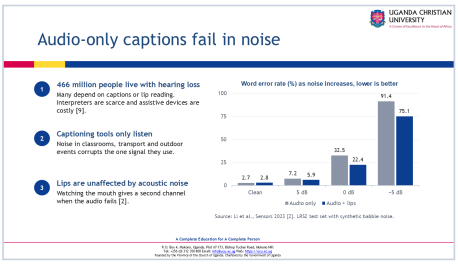
Group 4 proposal presentation. Eight minutes of talk, then about five minutes of questions. Times beside each slide add up to 8:00. Slides 11 (References) and 12 (Thank You) need no talking; keep slide 11 ready for questions about sources.



SLIDE 1 | 0:20

Title

- We are Group 4, presenting LipScribe.
- One line: LipScribe improves live captions in noisy places by watching the speaker's lips as well as listening.
- Outline: problem, user, data, pipeline, baselines, advanced components, evaluation, risks, six-week plan.



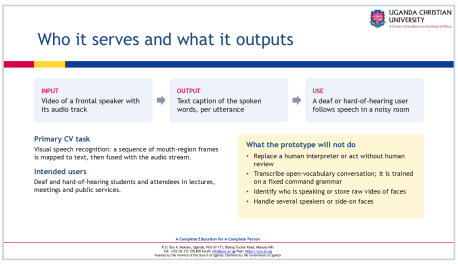
SLIDE 2 | 0:55

Audio-only captions fail in noise

- 466 million people live with hearing loss. Many rely on captions or lip reading, and interpreters and assistive devices are scarce or costly.
- Today's captioning tools only listen, so they break down in noisy classrooms, transport and outdoor events.
- Read the chart: audio-only goes from 2.7% word error rate in clean audio to 32.5% at 0 dB. Adding lips brings that to 22.4%.
- Lips are not affected by acoustic noise. That is the opening for computer vision.
- Be honest about minus 5 dB: both systems are still poor (91.4% and 75.1%), so we report every noise level.

If asked: *Why is this a computer-vision problem and not just a speech problem?*

The information that survives noise is visual. The core of the work is reading a video of the mouth, a sequence of frames mapped to text.

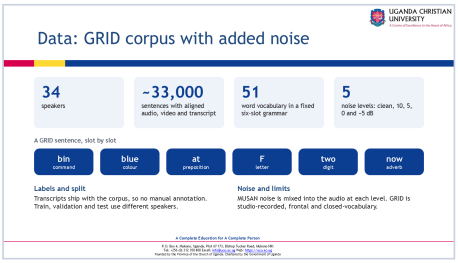


SLIDE 3 | 0:45

Who it serves and what it outputs

- User: a deaf or hard-of-hearing person following speech in a lecture, meeting or public service.
- Input: video of a frontal speaker with its audio. Output: a text caption per utterance.
- Use: the person can follow what is said in a noisy room.
- Primary CV task: visual speech recognition, video frames to text, fused with audio.
- Limits we state up front: it does not replace an interpreter, handle open conversation, identify speakers, or cope with several speakers or side-on faces.

If asked: *Why is sequence-to-text the right output for this user?*
The user needs the words. A class label or a detection box would not let them follow speech; a caption does.

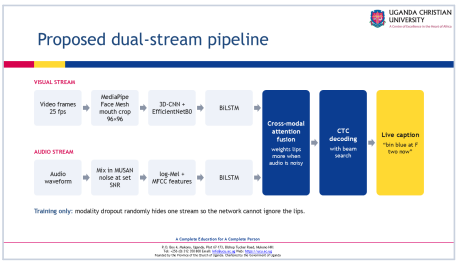


SLIDE 4 | 0:50

Data: GRID corpus with added noise

- GRID audiovisual corpus: 34 speakers, about 33,000 short sentences, clean frontal video with aligned audio.
- Every sentence follows a fixed six-slot grammar, for example "bin blue at F two now". The vocabulary is 51 words.
- Transcripts come with the corpus, so no manual annotation is needed.
- We mix MUSAN noise into the audio at five levels: clean, 10, 5, 0 and minus 5 dB.
- Split is by speaker, so test speakers are never seen in training.
- Limitation: studio-recorded, frontal, closed vocabulary. We come back to this under risks.

If asked: *Show evidence that the data are actually obtainable.*
GRID and MUSAN are public research corpora that can be downloaded now. Have a few real GRID frames or mouth crops ready to show.



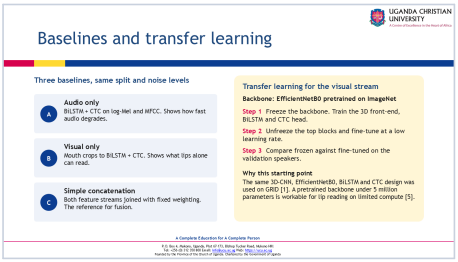
SLIDE 5 | 1:10

Proposed dual-stream pipeline

- Walk the visual stream left to right: frames, then MediaPipe Face Mesh finds the lips and we crop the mouth to 96 by 96.
- A 3D-CNN with an EfficientNetB0 backbone extracts lip shape and motion. A BiLSTM models the sequence in both directions.
- Audio stream: waveform with noise mixed in, converted to log-Mel and MFCC features, encoded by its own BiLSTM.
- Fusion: cross-modal attention gives the lips more weight when the audio is noisy.
- CTC decoding with beam search turns the fused sequence into text, without needing frame-level labels.
- Training only: modality dropout hides one stream at random so the network cannot ignore the lips.

If asked: *Why CTC?*

It maps a variable-length sequence of frames directly to text, so we do not need labels for each frame. GRID gives sentence transcripts, which is all CTC needs.



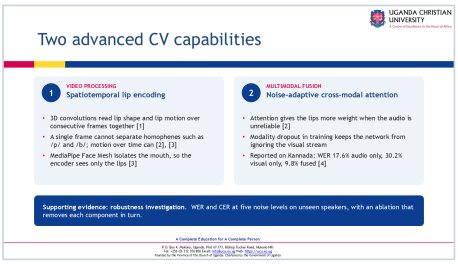
SLIDE 6 | 0:50

Baselines and transfer learning

- Three baselines on the same split and noise levels: audio only, visual only, and simple concatenation.
- Each answers a question: how fast audio degrades, what lips alone can read, and what naive fusion gives.
- Transfer learning: the visual stream starts from EfficientNetB0 pretrained on ImageNet.
- Step 1 freeze the backbone and train around it. Step 2 unfreeze the top blocks at a low learning rate. Step 3 compare the two.
- Why: LipSyncNet used this design on GRID, and a small pretrained backbone suits our limited compute.

If asked: *What will you do if the planned model does not improve over the baseline?*

Report it as a finding, use the error analysis to see where fusion fails, and fall back to the best baseline as the final system with the ablation as evidence.

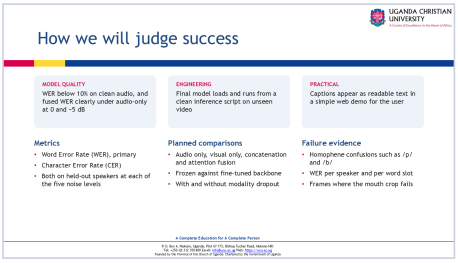


SLIDE 7 | 0:55

Two advanced CV capabilities

- Capability 1, video processing: 3D convolutions read lip shape and motion over neighbouring frames.
- Value: a single frame cannot separate homophenes like /p/ and /b/. Motion over time can.
- Capability 2, multimodal fusion: attention that adapts to noise, with modality dropout in training.
- Evidence from KanAVNet: 17.6% word error rate audio only, 30.2% visual only, 9.8% fused.
- Supporting both: a robustness investigation across five noise levels on unseen speakers, with an ablation that removes each component in turn.

If asked: *Why do these add value rather than complexity?*
Each targets a named failure: the 3D encoder targets homophenes, the fusion targets noisy audio. The ablation will show whether each one earns its place.

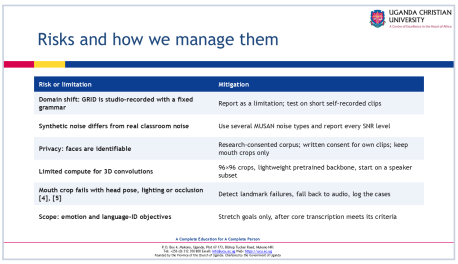


SLIDE 8 | 0:50

How we will judge success

- Primary metric: word error rate, with character error rate, on held-out speakers at every noise level.
- Model quality: below 10% word error rate on clean audio, and fused clearly better than audio only at 0 and minus 5 dB.
- Engineering: the final model loads and runs from a clean inference script on unseen video.
- Practical: captions appear as readable text in a simple web demo.
- Comparisons: the three baselines against attention fusion, frozen against fine-tuned, with and without modality dropout.
- Failure evidence: homophene confusions, error per speaker and per word slot, frames where the mouth crop fails.
- Targets may be revised after the baseline. Any change will be explained.

If asked: *What metric matters most and what does it measure?*
Word error rate: substitutions, deletions and insertions divided by the number of words in the reference. Lower is better.



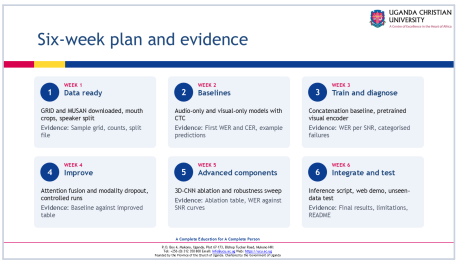
SLIDE 9 | 0:45

Risks and how we manage them

- Domain shift: GRID is studio-recorded with a fixed grammar. We state it as a limitation and test on short clips of our own.
- Synthetic noise is not real classroom noise. We use several noise types and report every level.
- Privacy: faces are identifiable. GRID was recorded for research; our own clips need written consent, and we keep mouth crops only.
- Compute: small crops, a lightweight pretrained backbone, and a speaker subset first.
- Mouth crop can fail with pose, lighting or occlusion. We detect it, fall back to audio and log the cases.
- Scope: emotion and language identification are stretch goals, only after core transcription meets its criteria.

If asked: *What is the most dangerous or misleading error your system could make?*

A fluent but wrong caption that the user trusts, for example a swapped letter or digit. That is why captions support the user and do not replace human review.



SLIDE 10 | 0:40

Six-week plan and evidence

- Week 1 data: GRID and MUSAN downloaded, mouth crops, speaker split. Evidence: sample grid, counts, split file.
- Week 2 baselines: audio-only and visual-only. Evidence: first WER and CER, example predictions.
- Week 3 train and diagnose: concatenation baseline and pretrained visual encoder. Evidence: WER per noise level, categorised failures.
- Week 4 improve: attention fusion and modality dropout. Evidence: baseline against improved table.
- Week 5 advanced components: 3D-CNN ablation and robustness sweep. Evidence: ablation table, WER against noise curves.
- Week 6 integrate and test: inference script, web demo, unseen-data test. Evidence: final results, limitations, README.
- Close: every member can explain every stage. Thank you, we welcome questions.

If asked: *How will another person reproduce your final inference pipeline?*

A repository with a README, a fixed split file, saved weights and one inference script that runs on a new video with no notebook state or personal paths.